Element I

Our most important criteria was that the rose petals fell at three specific times that could be controlled by the crew. We tested our method of dropping the petals which was the motors salvaged from the remote control cars. We originally have three motors that were controlled by a seperate controller. Once we had the motors in place on the base, we tried each controller to make sure each motors functioned and could spin both directions. We also test how fast the motor spun by attaching the fishing line to the motor and measuring how long it would take them to spool up 1 meter of line. After we rewired the motors to function using only one controller we repeteded the same test and obtained the same results.

The second most important criteria was that there must be three lights pointed up at the rose. We made sure the lights we bought turned on using the batteries we had. Once we knew they functioned, we attached them in place. We set the rose up on the stage where it would be on stage and turned the lights on to the brightest setting. We then had a group members sit in the front row, a row about half way back in the theater, and in the very back row. That group member rated on a scale of 1-10 on how well they could see the light and how well it illuminated the rose.

We also tested the safety and reliability of the rose in one test. For the test we dropped one of the petals 10 times in a row after winding them back up each time. After each trial we recorded any mechanical or visual problems. We also recorded which part of the rose had the problem. After the 10 trials, we fixed the problems we noticed and repeated the 10 trials. We averaged the amount of problems we noticed on the 2 tests. This showed us how reliable our product was and allowed us to find which parts of the mechanism broke most often. This allowed us to pin point which parts we needed to change or refine to make the product reliable for the use by the theater crew and cast.